

REEP — Technology Stack & Interaction Map

Which technologies are in play, and exactly what travels between them

Angular 22.1 · TypeScript 6.0 · FastAPI 0.141 · Python 3.14 · SQLAlchemy 2.0 · PostgreSQL 17

61 tables · 44 migrations · 19 student routes · 34 lazy chunks · 609 backend tests

Print A3 landscape (420 × 297 mm) at 100 % scale, no margins

A · CLIENT RUNTIME — the browser takes

ANGULAR 22.1 · TYPESCRIPT 6.0

Standalone components — no NgModules anywhere
Signals hold state; RxJS 7.8 only where HttpClient needs it
 Router: every route is loadComponent() → its own chunk
 ReactiveForms + FormsModule(ngModel) per screen
 app-shell picks one of three navs from the session role
 apps/web/src/app/{core,layout,features}

SCREEN MAP

```
/student/* — 19 routes (badges, ledger, jobs, resume, ...)
/mentor/* — Faculty (MENTOR): mentees · leave · upskilling
/admin/* — Main Admin (ADMIN): setup · people · exports
/alumni · /alumni/jobs · /login · /register
```

PLATFORM APIs THE SCREENS USE

- fetch(url, {credentials:'include'}) — the house pattern
- FormData for uploads (certificates, resumes, photos)
- WebSocket + AudioWorklet — the live mock interview
- MediaDevices.getUserMedia → 24 kHz mono PCM16
- Blob + anchor for PDF reports and CSV exports
- No token in JS: the session cookie is httpOnly

DESIGN SYSTEM & ASSETS

reep-v2.scss — global classes (.card .chip .dt-table)
One committed theme · status is text + colour, never colour
Self-hosted fonts; Material Symbols SUBSET (144 kB not 5.2 MB)
.icon stays hidden until FontFace reports itself loaded
apexcharts / ng-apexcharts on the analytics screens

BUILD & CLIENT TELEMETRY

@angular/build 22.1 (esbuild) → dist/web/browser
172 kB initial · 34 lazy chunks · budgets enforced in CI
Static files are served by CloudFront + S3, not by the API
@sentry/angular 10.71 — dynamic import, only when a DSN is se
vite 4 + jsdom 28 for client unit tests

WHAT THE CLIENT NEVER DOES

- Never reads the session cookie — it cannot
- Never calls a model directly; all AI goes through /api
- Never holds another student's row — scoping is server-side
- Its role check drives navigation only, never authorisation

RETAINED IN THE CLIENT

livekit-client 2.21 — the voice rollback path, no UI calle

B · API PROCESS — one request, top to bottom

1. UVICORN 0.52.3 — ASGI SERVER

- uvloop + httptools · HTTP/1.1 and WebSocket on :3300
- Sync `def` endpoints run in a 40-thread pool — which is what**
- keeps the event loop free for live interview audio

2. STARLETTE MIDDLEWARE — IN ORDER

CORSMiddleware — one explicit origin, credentials allowed
 RequestTraceMiddleware — mints or keeps X-Request-ID, tags
 Sentry, echoes the header, prints one access line

3. FASTAPI 0.141.1 — ROUTING & D

- One APIRouter per domain, all mounted under /api
- Depends(get_current_session) on every protected route
- response_model=... ⇒ the response shape is the contract

4. IDENTITY & SESSION

PyJWT 2.13 HS256 verify over the reep_session cookie
Claims: userId · email · name · role · studentId? · mentorId?
token_version compared on the way in ⇒ logout can revoke
Password path (dev/CI only): scrypt N=16384 r=8 p=1 dklen=64

5. SCOPE GATES — THE TWO RULES, IN CODE

`_require_student · require_mentor · require_admin`
`_assert_can_access_student` — ONE implementation, imported
 Out-of-scope ids flatten to 404, never 403 (no membership oracle)

6. PYDANTIC 2.13 — VALIDATION & SETTINGS

Bodies parsed, coerced and rejected at the edge
pydantic-settings 2.15 loads .env into one Settings object
Settings also owns the production boot refusal

7. SERVICE LAYER — WHERE THE WORK HAPPENS

document_store · ai/llm · knowledge · conversations · retention
interview_relay · interview_audio · redaction · ratelimit
resume_pdf / english_report — ReportLab 5.0, strictly local

8. PERSISTENCE — SQLALCHEMY 2.0.52

Session per request via Depends(get_db)
2.0-style select(); models are the schema's source of truth
psycopg 3.3.4 binary protocol · pool_pre_ping, sized by settings
Alembic 1.19 owns DDL — one head, hand-written enum steps

BUILD, TEST & CI — FOUR JOBS

```

pytest — 609 tests across 37 modules; TestClient boots the SAME ASGI app
REEP_REQUIRE_DB=1 turns a missing Postgres into a hard failure, not a green skip
api-imports — proves every app/ module imports against requirements.txt ALONE
worker-imports — the same proof for the voice worker's separate manifest
web — ng build, where a bundle budget fails a route that stopping being lazy
viteest 4 + jsdom 28 for client unit tests + prettier for format

```

PACKAGING & PROCESS TOPOLOGY

Dockerfile: python:3.14-slim-bookworm, dependency layer before code requirements.txt is runtime-only and pinned == ; -dev.txt adds pytest
Voice worker needs its OWN venv on Python 3.12 – livekit requires < 3.11
Processes: the API (uvicorn) · a scheduled retention task · Postgres
...and optionally the voice worker. The AI interviewer runs INSIDE the API.
Terraform provisions the infrastructure; Alembic owns the schema

THE GUARDS THAT RIDE THESE WIRES

RULE 1 — wires 7 & 8: student PII may not reach a remote model unless the egress flag is open; refused in-process, and the resume composes deterministically

RULE 2 — wire 11: staff scope narrowed in SQL, never filtered afterwards

CONSENT — wire 6: no live grant and the socket never opens (close 4013)

TRACE — every wire: one X-Request-ID, echoed, tagged on Sentry, printed as rid=

BOOT GUARD — the process refuses to start on repo-default secrets in production

Numbered discs key each wire to its protocol · payload · guard. A dashed return arrow means traffic flows both ways on the same connection.

Generated by tools/diagrams/render_stack_interaction.py — regenerate when the stack changes